Fitting Models by Bayesian Inference with JAGS

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This page fits models by Bayesian inference, using the JAGS sampler driven from R: a single proportion, a logistic regression, a survival model, and a random-effects model, and then averages over linear regression models (Dobson and Barnett 2018, chap. 14). It uses the priors of the Bayesian Inference page and the sampling and convergence checks of the Markov Chain Monte Carlo page.

A first example: a single proportion

From here on, the models are fit by Bayesian inference with JAGS ("Just Another Gibbs Sampler"), a program that builds an MCMC sampler from a text description of a model (Dobson and Barnett 2018, chap. 13), driven from R through the `rjags` package. We begin with the simplest possible model, a single Bernoulli probability, whose exact posterior is known from the Beta-Bernoulli example, so the output can be checked. The example shows the mechanics that every later analysis reuses: specifying a model, supplying data, running a burn-in, monitoring parameters, and summarizing and checking the draws. In JAGS, `dnorm(mean, precision)` is parameterized by the precision, the reciprocal of the variance.

Example 0.1 (A Bernoulli model fitted with JAGS)

Example 0.1 (A Bernoulli model fitted with JAGS)**.** The data are $r = 55$ successes in $n = 91$ trials, and the prior is uniform, as in the Beta-Bernoulli example. Each chain gets its own random-number generator seed, so the output is reproducible:

```r
y <- rep(c(1L, 0L), times = c(55L, 36L))
bernoulli_model <- "
model {
  for (i in 1:N) {
    y[i] ~ dbern(p)
  }
  p ~ dbeta(1, 1)
}
"
inits <- list(
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 1),
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 2)
)
fit <- rjags::jags.model(
  textConnection(bernoulli_model),
  data = list(y = y, N = length(y)),
  inits = inits,
  n.chains = 2,
  quiet = TRUE
)
stats::update(fit, n.iter = 1000, progress.bar = "none") # burn-in
draws <- rjags::coda.samples(
  fit,
  variable.names = "p",
  n.iter = 4000,
  progress.bar = "none"
)
p_draws <- as.matrix(draws)[, "p"]
c(
  mean = mean(p_draws),
  stats::quantile(p_draws, c(0.025, 0.975)),
  Rhat = coda::gelman.diag(draws)$psrf[1, "Point est."]
) |>
  round(3)
#>  mean  2.5% 97.5%  Rhat
#> 0.601 0.499 0.698 1.000
```

The posterior mean and equal-tailed 95% credible interval agree with the exact Beta(56, 37) values of the Beta-Bernoulli example to about two decimal places, and the potential scale reduction factor is 1.

Binary outcomes: logistic regression

For a binary outcome $Y_i \sim \text{Bernoulli}(\pi_i)$ with $\text{logit}(\pi_i) = \tilde{x}_i \cdot \tilde{\beta}$, the logistic regression model, a Bayesian analysis places a prior on the coefficient vector $\tilde{\beta}$ and samples the posterior $\text{p}(\tilde{\beta} \mid \tilde{y})$ by MCMC (Dobson and Barnett 2018, chap. 14, p. 318). Each β_{x_j} is summarized by the mean and quantiles of its draws. No large-sample Gaussian approximation is needed: a credible interval is read directly from the posterior quantiles, and the posterior of an odds ratio $e^{\beta_{x_j}}$, or of any other function of $\tilde{\beta}$, is obtained by transforming the draws. For a video introduction, see Richard McElreath's lecture *Modeling Events* (Statistical Rethinking 2026, Lecture A09).

Example 0.2 (Logistic regression fitted by Bayesian inference)

Example 0.2 (Logistic regression fitted by Bayesian inference)**.** We simulate $N = 200$ observations from a logistic model with one continuous predictor, intercept $\beta_0 = -0.5$ and slope $\beta_x = 1.2$, and place a diffuse $\text{N}(0, 10^2)$ prior on each coefficient (precision 0.01 in JAGS). The JAGS program also monitors the odds ratio e^{β_x} for a one-unit increase in x. In the code, `beta0` and `beta1` are β_0 and β_x.

```r
set.seed(2024)
n_obs <- 200L
x <- stats::rnorm(n_obs)
y <- stats::rbinom(n_obs, size = 1, prob = stats::plogis(-0.5 + 1.2 * x))

logistic_model <- "
model {
  for (i in 1:N) {
    y[i] ~ dbern(pi[i])
    logit(pi[i]) <- beta0 + beta1 * x[i]
  }
  beta0 ~ dnorm(0, 0.01)
  beta1 ~ dnorm(0, 0.01)
  OR <- exp(beta1)
}
"
inits <- list(
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 1),
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 2)
)
fit <- rjags::jags.model(
  textConnection(logistic_model),
  data = list(y = y, x = x, N = n_obs),
  inits = inits,
  n.chains = 2,
  quiet = TRUE
)
stats::update(fit, n.iter = 1000, progress.bar = "none")
draws <- rjags::coda.samples(
  fit,
  variable.names = c("beta0", "beta1", "OR"),
  n.iter = 4000,
  progress.bar = "none"
)
draws_matrix <- as.matrix(draws)
cbind(
  mean = colMeans(draws_matrix),
  t(apply(draws_matrix, 2, stats::quantile, probs = c(0.025, 0.975))),
  Rhat = coda::gelman.diag(draws, multivariate = FALSE)$psrf[, "Point est."]
) |>
  round(3)
#>         mean   2.5%  97.5%  Rhat
#> OR     3.417  2.275  5.132 1.000
#> beta0 -0.654 -1.002 -0.316 1.001
#> beta1  1.207  0.822  1.635 1.000
```

Each 95% credible interval contains its coefficient's true value, and every $\hat{R}$ is near 1. As comparing MCMC estimates to maximum likelihood leads us to expect, the posterior means are close to the maximum likelihood estimates:

```
stats::glm(y ~ x, family = stats::binomial()) |>
  stats::coef() |>
  round(3)
#> (Intercept)           x
#>      -0.647       1.183
```

Survival analysis

Parametric survival models, such as those with exponential or Weibull event times, can be fit by Bayesian inference: priors are placed on the baseline-hazard parameters and the regression coefficients, and the posterior is sampled by MCMC (Dobson and Barnett 2018, chap. 14, p. 327). Censoring enters through the likelihood, exactly as in maximum likelihood estimation with the likelihood with censoring: an observed event at time t contributes its density $\lambda(t)\ \mathrm{S}(t)$, and an observation right-censored at time t contributes its survival probability $\mathrm{S}(t)$, where λ is the hazard and S the survival function.

When a model's log-likelihood contribution ℓ_i for observation i is not one of the distributions built into JAGS, the *zeros trick* supplies it. An observed value of 0 from a Poisson distribution with mean ϕ_i has probability $e^{-\phi_i}$, and with $\phi_i \stackrel{\text{def}}{=} C - \ell_i$ for a constant C,

$$\begin{aligned} e^{-\phi_i} &= e^{-(C-\ell_i)} && \text{(definition of } \phi_i\text{)} \\ &= e^{-C} e^{\ell_i} && \text{(splitting the exponent)} \\ &\propto e^{\ell_i} && (C \text{ does not depend on the parameters}), \end{aligned}$$

which is observation i's likelihood contribution. So declaring data `zeros[i] = 0` with `zeros[i] ~ dpois(C - loglik[i])` multiplies the likelihood by e^{ℓ_i}. The constant C only has to be large enough to keep every ϕ_i positive.

Example 0.3 (Exponential survival regression fitted by Bayesian inference)

Example 0.3 (Exponential survival regression fitted by Bayesian inference)**.** We simulate $N = 200$ right-censored exponential survival times, with a binary covariate x_i (say, treatment) and hazard $\lambda_i = \exp\{\beta_0 + \beta_x x_i\}$, where the log baseline hazard is $\beta_0 = \log 0.05$ and the log hazard ratio is $\beta_x = -0.7$. With event indicator δ_i (1 for an event, 0 for a censored time) and follow-up time t_i, the exponential density is $\lambda_i e^{-\lambda_i t}$ and its survival function is $e^{-\lambda_i t}$, so observation i contributes the log-likelihood

$$\ell_i \stackrel{\text{def}}{=} \delta_i \log \lambda_i - \lambda_i t_i.$$

JAGS has no built-in distribution for this contribution, so we fit it with the zeros trick. In the code, `beta0` and `beta1` are β_0 and β_x.

```r
set.seed(2026)
n_obs <- 200L
x <- stats::rbinom(n_obs, size = 1, prob = 0.5)
event_time <- stats::rexp(n_obs, rate = exp(log(0.05) - 0.7 * x))
censor_time <- stats::rexp(n_obs, rate = 0.03)
follow_up <- pmin(event_time, censor_time)
event <- as.integer(event_time <= censor_time)

survival_model <- "
model {
  C <- 10000
  for (i in 1:N) {
    log(lambda[i]) <- beta0 + beta1 * x[i]
    loglik[i] <- event[i] * log(lambda[i]) - lambda[i] * follow_up[i]
    zeros[i] ~ dpois(C - loglik[i])
  }
  beta0 ~ dnorm(0, 0.01)
  beta1 ~ dnorm(0, 0.01)
  HR <- exp(beta1)
}
"
inits <- list(
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 1),
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 2)
)
fit <- rjags::jags.model(
  textConnection(survival_model),
  data = list(
    follow_up = follow_up, x = x, event = event,
    N = n_obs, zeros = rep(0, n_obs)
  ),
  inits = inits,
  n.chains = 2,
  quiet = TRUE
)
stats::update(fit, n.iter = 1000, progress.bar = "none")
draws <- rjags::coda.samples(
  fit,
  variable.names = c("beta0", "beta1", "HR"),
  n.iter = 4000,
  progress.bar = "none"
)
draws_matrix <- as.matrix(draws)
cbind(
  mean = colMeans(draws_matrix),
  t(apply(draws_matrix, 2, stats::quantile, probs = c(0.025, 0.975))),
  Rhat = coda::gelman.diag(draws, multivariate = FALSE)$psrf[, "Point est."]
) |>
  round(3)
#>          mean   2.5%  97.5% Rhat
#> HR      0.602  0.390  0.875    1
#> beta0 -3.119 -3.391 -2.870    1
#> beta1 -0.520 -0.941 -0.133    1
```

About 50% of the follow-up times end in an event. The 95% credible interval for β_x contains the true -0.7, though the posterior mean is some distance from it. The maximum likelihood estimates, from the same log-likelihood, show that the gap is sampling variation in this data set rather than an effect of the prior:

```
neg_loglik <- function(beta) {
  lambda <- exp(beta[1] + beta[2] * x)
  -sum(event * log(lambda) - lambda * follow_up)
}
stats::optim(c(0, 0), neg_loglik)$par |>
  stats::setNames(c("beta0", "beta1")) |>
  round(3)
#>  beta0  beta1
#> -3.112 -0.524
```

The monitored e^{β_x} gives the hazard ratio and its credible interval directly.

Random effects

The hierarchical model of the two-level Gaussian example is the random-effects *model structure*, with group-level parameters $\theta_j \sim \mathrm{N}(\mu, \tau^2)$ drawn from a common distribution. What changes here from a maximum likelihood fit is the *inference method*: the Bayesian analysis places priors on the hyperparameters μ and τ and samples their joint posterior together with the group-level parameters (Dobson and Barnett 2018, chap. 14, p. 329). The posterior shrinks each group's estimate toward the overall mean, by an amount the data determine through τ, and the posterior for τ carries the uncertainty about the between-group spread into every group-level summary.

Example 0.4 (A random-intercept model fitted by Bayesian inference)

Example 0.4 (A random-intercept model fitted by Bayesian inference)**.** We simulate $J = 8$ groups of 12 observations each, with group means drawn from $\mathrm{N}(\mu, \tau^2)$ ($\mu = 5$, $\tau = 1.5$) and within-group standard deviation $\sigma = 2$. The priors are a diffuse $\mathrm{N}(0, 100^2)$ for μ and flat priors on $(0, 100)$ for the standard deviations τ and σ. Being flat on the standard-deviation scale is a choice: as a flat prior on the log-odds shows for a probability, it is not flat on another scale, such as the variance.

```r
set.seed(2025)
n_groups <- 8L
n_per_group <- 12L
theta_true <- stats::rnorm(n_groups, mean = 5, sd = 1.5)
group <- rep(seq_len(n_groups), each = n_per_group)
y <- stats::rnorm(n_groups * n_per_group, mean = theta_true[group], sd = 2)

random_intercept_model <- "
model {
  for (i in 1:N) {
    y[i] ~ dnorm(theta[group[i]], 1 / sigma^2)
  }
  for (j in 1:J) {
    theta[j] ~ dnorm(mu, 1 / tau^2)
  }
  mu ~ dnorm(0, 1.0E-4)
  sigma ~ dunif(0, 100)
  tau ~ dunif(0, 100)
}
"
inits <- list(
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 1),
  list(.RNG.name = "base::Mersenne-Twister", .RNG.seed = 2)
)
fit <- rjags::jags.model(
  textConnection(random_intercept_model),
  data = list(y = y, group = group, N = length(y), J = n_groups),
  inits = inits,
  n.chains = 2,
  quiet = TRUE
)
stats::update(fit, n.iter = 2000, progress.bar = "none")
draws <- rjags::coda.samples(
  fit,
  variable.names = c("mu", "tau", "sigma", "theta"),
  n.iter = 10000,
  progress.bar = "none"
)
draws_matrix <- as.matrix(draws)
posterior_summary <- cbind(
  mean = colMeans(draws_matrix),
  t(apply(draws_matrix, 2, stats::quantile, probs = c(0.025, 0.975))),
  Rhat = coda::gelman.diag(draws, multivariate = FALSE)$psrf[, "Point est."]
)
round(posterior_summary[c("mu", "tau", "sigma"), ], 3)
#>        mean  2.5% 97.5%  Rhat
#> mu    5.572 4.733 6.405 1.002
#> tau   0.868 0.125 2.002 1.004
#> sigma 2.161 1.867 2.520 1.000
```

The 95% credible intervals contain the true $\mu = 5$, $\tau = 1.5$ and $\sigma = 2$. The interval for τ is wide: eight groups say little about how spread out group means are, and the posterior reports that uncertainty instead of a single estimate of the variance component.
Each group's posterior mean lies between its sample mean and the overall sample mean:

```
sample_means <- as.vector(tapply(y, group, mean))
tibble::tibble(
  group_id = seq_len(n_groups),
  sample_mean = sample_means,
  posterior_mean = posterior_summary[
    paste0("theta[", seq_len(n_groups), "]"), "mean"
  ]
) |>
  dplyr::mutate(overall_mean = mean(y)) |>
  round(2)
#> # A tibble: 8 x 4
#>   group_id sample_mean posterior_mean overall_mean
#>      <dbl>       <dbl>          <dbl>        <dbl>
#> 1        1        6.08           5.87         5.56
#> 2        2        5.53           5.55         5.56
#> 3        3        6.14           5.9          5.56
#> 4        4        7.27           6.57         5.56
#> 5        5        5.1            5.3          5.56
#> 6        6        4.31           4.85         5.56
#> 7        7        5.51           5.53         5.56
#> 8        8        4.56           4.99         5.56
```

This is the borrowing of strength described in distributions and hierarchies: group 4, with the largest sample mean, is pulled furthest toward the overall mean.

Bayesian model averaging

When several candidate models are plausible, committing to a single "best" one ignores the uncertainty about which model is right. Bayesian model averaging instead averages over the models, weighting each by its posterior probability (Dobson and Barnett 2018, chap. 14, p. 339). This approach carries *model* uncertainty, not just parameter uncertainty, into the final inference. It is an alternative to choosing a single model by predictor selection.

Definition 0.1 (Posterior model probability)

Definition 0.1 (Posterior model probability)**.** Let $M_1, \ldots, M_K$ be candidate models with

prior probabilities $\mathrm{p}(M_k)$ summing to 1, and let $\mathrm{p}(\tilde{y} \mid M_k)$ be the marginal likelihood of the data under model M_k. The **posterior model probability** of M_k is

$$\mathrm{p}(M_k \mid \tilde{y}) \stackrel{\text{def}}{=} \frac{\mathrm{p}(\tilde{y} \mid M_k)\ \mathrm{p}(M_k)}{\sum_{l=1}^{K} \mathrm{p}(\tilde{y} \mid M_l)\ \mathrm{p}(M_l)}.$$

Example 0.5 (Posterior probabilities of two models)

Example 0.5 (Posterior probabilities of two models)**.** Two models with equal prior probabilities

$$\begin{aligned} \mathrm{p}(M_1) &= \mathrm{p}(M_2) \\ &= 0.5 \end{aligned}$$

and marginal likelihoods $\mathrm{p}(\tilde{y} \mid M_1) = 0.02$ and $\mathrm{p}(\tilde{y} \mid M_2) = 0.06$ have posterior probabilities

$$\begin{aligned} \mathrm{p}(M_1 \mid \tilde{y}) &= 0.01/(0.01 + 0.03) \\ &= 0.25 \end{aligned}$$

and $\mathrm{p}(M_2 \mid \tilde{y}) = 0.75$.

Definition 0.2 (Bayesian model averaging)

Definition 0.2 (Bayesian model averaging)**.** With the posterior model probabilities $\mathrm{p}(M_k \mid \tilde{y})$ of candidate models $M_1, \ldots, M_K$, **Bayesian model averaging** estimates a quantity Δ that has the same meaning in every model by its posterior distribution averaged over the models:

$$\mathrm{p}(\Delta \mid \tilde{y}) \stackrel{\text{def}}{=} \sum_{k=1}^{K} \mathrm{p}(\Delta \mid M_k, \tilde{y})\ \mathrm{p}(M_k \mid \tilde{y}).$$

Definition 0.3 (Posterior inclusion probability)

Definition 0.3 (Posterior inclusion probability)**.** When the candidate models of Definition 0.2 differ in which predictors they include, the **posterior inclusion probability** of a predictor is the total posterior probability of the models that include it.

Example 0.6 (A BIC approximation to Bayesian model averaging)

Example 0.6 (A BIC approximation to Bayesian model averaging)**.** Marginal likelihoods

are hard to compute, but with equal prior probabilities for the models, the Bayesian information criterion gives a large-sample approximation to the posterior model probabilities (Schwarz 1978):

$$\mathrm{p}(M_k \mid \tilde{y}) \approx \frac{\exp\{-\frac{1}{2}\,\mathrm{BIC}_k\}}{\sum_{l=1}^{K} \exp\{-\frac{1}{2}\,\mathrm{BIC}_l\}}.$$

We simulate data in which only x_1 and x_2 affect the outcome, fit a linear regression for every subset of the three predictors x_1, x_2 and x_3, and compute these approximate posterior model probabilities:

```r
set.seed(2027)
n_obs <- 120L
dat <- tibble::tibble(
  x1 = stats::rnorm(n_obs),
  x2 = stats::rnorm(n_obs),
  x3 = stats::rnorm(n_obs)
) |>
  dplyr::mutate(y = 1 + 0.8 * x1 + 0.5 * x2 + stats::rnorm(n_obs))

predictors <- c("x1", "x2", "x3")
subsets <- lapply(0:3, function(k) {
  utils::combn(predictors, k, simplify = FALSE)
}) |>
  unlist(recursive = FALSE)
models <- lapply(subsets, function(vars) {
  rhs <- if (length(vars) > 0) paste(vars, collapse = " + ") else "1"
  stats::lm(stats::as.formula(paste("y ~", rhs)), data = dat)
})
bic <- vapply(models, stats::BIC, numeric(1))
weight <- exp(-0.5 * (bic - min(bic)))
weight <- weight / sum(weight)

tibble::tibble(
  model = vapply(
    subsets,
    function(s) if (length(s) > 0) paste(s, collapse = " + ") else "(none)",
    character(1)
  ),
  weight = round(weight, 3)
)
#> # A tibble: 8 x 2
#>   model        weight
#>   <chr>         <dbl>
#> 1 (none)        0
#> 2 x1            0
#> 3 x2            0
#> 4 x3            0
#> 5 x1 + x2       0.848
#> 6 x1 + x3       0
#> 7 x2 + x3       0
#> 8 x1 + x2 + x3  0.152
```

Subtracting the smallest BIC before exponentiating leaves the normalized weights unchanged and avoids numerical underflow. The posterior inclusion probability of each predictor sums the weights of the models that contain it:

```
includes <- function(v) vapply(subsets, function(s) v %in% s, logical(1))
vapply(predictors, function(v) sum(weight[includes(v)]), numeric(1)) |>
  round(3)
#>    x1    x2    x3
#> 1.000 1.000 0.152
```

For the model-averaged slope of x_1, we approximate its posterior mean within each model by the least-squares estimate, taken as 0 in models that exclude x_1, and average with the posterior model probabilities as weights:

```
beta_x1 <- vapply(models, function(m) {
  cf <- stats::coef(m)
  if ("x1" %in% names(cf)) cf[["x1"]] else 0
}, numeric(1))
c(
  bma_slope_x1 = sum(weight * beta_x1),
  least_squares_x1_x2 = stats::coef(stats::lm(y ~ x1 + x2, data = dat))[["x1"]]
) |>
  round(3)
#>        bma_slope_x1 least_squares_x1_x2
#>               0.947               0.948
```

The predictors that affect the outcome, x_1 and x_2, have inclusion probabilities near 1, and the irrelevant x_3 a much smaller one. The models without x_1 get almost no weight, so the model-averaged slope of x_1 is close to its least-squares estimate in the model with x_1 and x_2 only. Both differ from the true slope 0.8 by sampling variation in this data set, not because of the averaging.

References

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. https://doi.org/10.1201/9781315182780.

Schwarz, Gideon. 1978. "Estimating the Dimension of a Model." *The Annals of Statistics* 6 (2): 461–64. https://doi.org/10.1214/aos/1176344136.